



University of
Nottingham

UK | CHINA | MALAYSIA

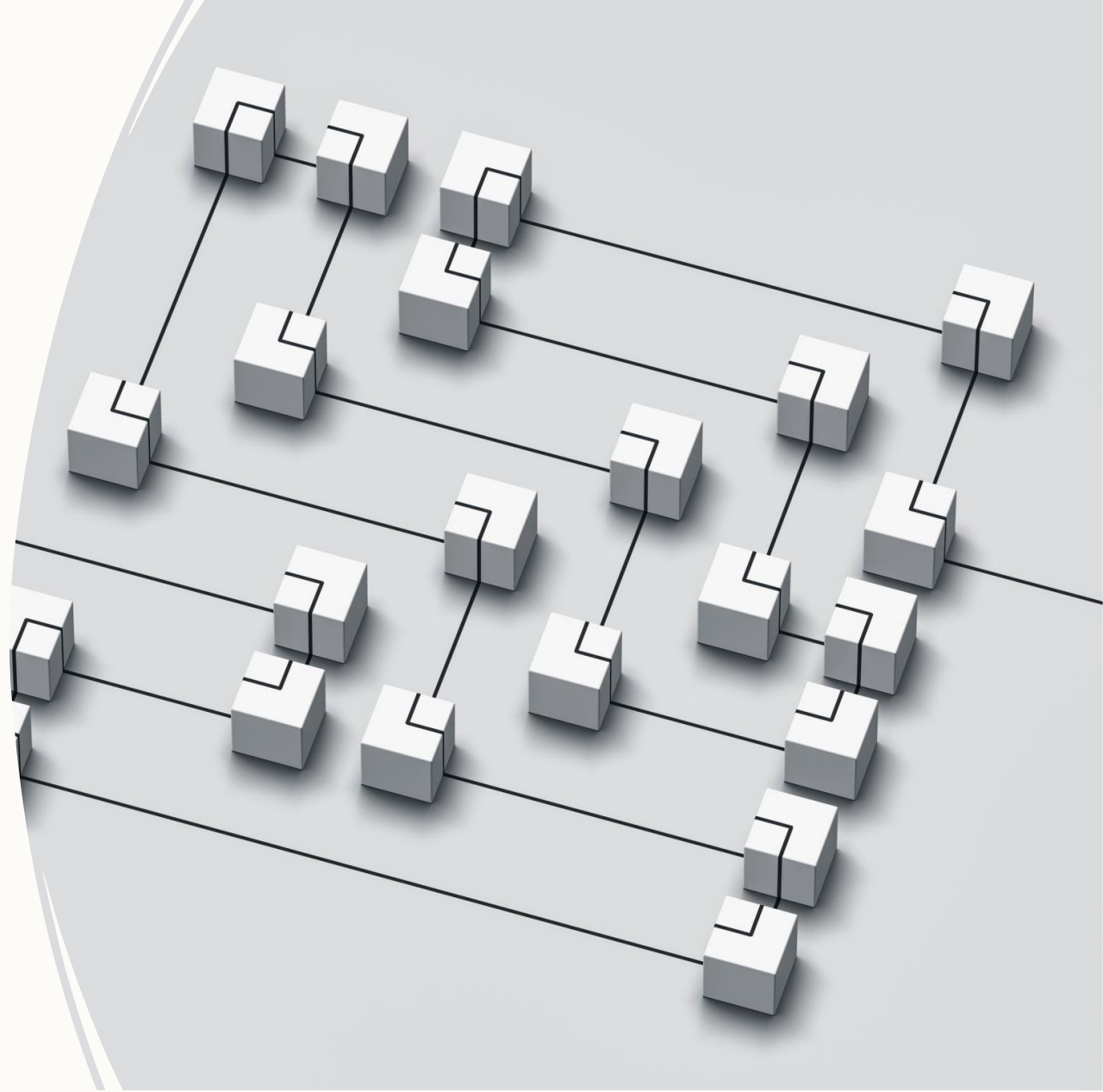
COMP 3056 Professional Ethics in Computing

Week 3 Ethics in Computing

Anthony Bellotti

Learning outcomes

- Why Ethics is important in Computing
- Impact of Computing
- Ethical Principles in Computing
- Ethical dilemmas and critical reasoning
- Doing the “right” thing





University of
Nottingham

UK | CHINA | MALAYSIA

Do we need Ethics in Computing?

“Why can’t I just get on and do my coding?!”

Two examples from UK (*but examples can be found in all countries!*)





Example 1: Horizon IT Scandal

- The Horizon IT System was introduced as a Post Office accounting system in UK in 1996.
- Over two decades bugs in the system caused accounting failures which were blamed on Post Office staff (sub-postmasters), many of whom were prosecuted for theft.
- The Post Office denied there were bugs but recent evidence suggested they were aware.

[Horizon scandal factsheet \(UK government\)](#)

<https://www.youtube.com/watch?v=3uIEBOBSzbk>

- What responsibility did the Post Office have?
- What responsibility does the IT supplier, Fujitsu, have?
- What responsibility did the software developers have?

Technical details of the bug: <https://www.youtube.com/watch?v=hBJm9ZYqL10&t=3s>





Example 2: Online content and children

- Molly Russell, a 14 year old girl took her own life in 2017.
- She had viewed extensive content related to suicide, depression and anxiety online.
- The coroner concluded that Molly died from "an act of self-harm while suffering from depression and the negative effects of online content".

[Molly Russell: Friend of 14-year-old who died from self-harm speaks out over Online Safety Bill](#)

<https://www.youtube.com/watch?v=AGPff4AApLA>



- Do social media companies have a responsibility?
- Do we need ethical considerations when introducing online and directed content?



Impact of Computing

- Safety critical systems such as airplane control systems have a direct impact on human experience. 飞机控制系统等安全关键系统直接影响人类体验。
- But any computer system can have a negative impact on people: even an accounting system. 但任何计算机系统都可能对人们产生负面影响：即使是会计系统。
- Ethics is needed to ensure that we deploy computer systems that are a positive experience for people and not harmful. 我们需要伦理道德来确保我们部署的计算机系统能为人们带来积极的体验，而不是有害的。
- Software developers and computer engineers have a responsibility to ensure ethics in computing. 软件开发人员和计算机工程师有责任确保计算领域的伦理道德

理论

在本页的体现

功利主义 (Utilitarianism)

系统设计要最大化总体幸福、最小化风险与伤害。

义务论 (Deontology)

工程师应遵守“不伤害”与“诚实透明”的基本道德义务。

契约论 (Contractarianism)

技术开发是社会契约的一部分：公众信任 → 专业群体应维护公共利益。

关怀伦理 (Ethics of Care)

工程师应考虑系统对用户的真实体验、心理健康、可获得性。



Some common ethical principles in Computing

计算领域的常见伦理原则

1. Contribute positively to society and human well-being. 积极促进社会与人类福祉
2. Be honest, transparent and trustworthy. 诚实、透明、值得信赖。
3. Maintain professional competence for yourself and other coworkers. 保持自身与同事的专业能力
4. Respect data privacy and intellectual property rights. 尊重数据隐私与知识产权。
5. Be fair, do not discriminate, and take action against discrimination. 公平公正，不歧视，并积极反歧视

Which of these do you think were broken in the two examples in the previous slides?

Note: *this is not an exhaustive list; can you think of others?*



Ethical dilemmas and critical reasoning

在面对伦理困境时，你要**主动做判断** (use judgement)

- Use your judgement, and reinforce with theories of ethics.
- If you say something is ethical, what is your reason?
不能只说“我觉得没问题”，要能逻辑推理、给出依据
- Use professional codes of conduct, as source of guidance:-
 1. ACM/IEEE Software Engineering Code of Ethics and Practice
 2. British Computer Society (BCS) Code of Conduct

计算机伦理规范很有价值，因为它们处理了你在计算领域可能遇到的伦理问题。

- Codes of Ethics in Computing are valuable since they deal with ethical issues you may encounter specifically in computing.

- In this module, we will focus on the BCS Code of Conduct. Key principles:-

- | | |
|-------------|--|
| 1. 公共利益 | 1. Public interest |
| 2. 专业能力和诚信 | 2. Professional competence and integrity |
| 3. 对相关部门的职责 | 3. Duty to relevant authority |
| 4. 对职业的职责 | 4. Duty to the profession |

• 一般结构:

- 事实描述 (what happened?)
- 涉及的人 (who is affected?)
- 使用哪种伦理理论 (which principle?)
- 结果推演 (what consequence?)
- 最终结论 (ethical / unethical)





Why don't we always do the “right” thing?

无知 (ignorance) 或失误 (mistake)

- We may do the wrong thing by **mistake or ignorance**; e.g. introduce a bug.

Forgivable error ” (可原谅的错误)

- But consider: at what point is this a “forgivable error” or “negligence”? Negligence ” (疏忽)

- Reminder from Buddha: **the source of all suffering is ignorance.** 佛教观点：无知 (ignorance) 是痛苦的根源

并非所有错误都是“无意的”；有时是明知故犯 (intentional misconduct) 。

- We may do the wrong thing because we are **intentionally bad.**
 - This may happen more than we might imagine!
 - Survey of IT professionals suggests that **subversion occurs on 20% of projects**, and **lying on 50% of projects.**

The Dark Side of Software Engineering, Rost & Glass (Wiley) 2011

20% 的项目出现了 “subversion (内部破坏或抵抗)” ；
50% 的项目中存在 “lying (撒谎)” ，例如伪报进度、隐藏风险、夸大成就。





Why don't we always do the “right” thing?

相对主义

- **Relativism**. We may not agree on what is “right”. 不同个体、文化或背景对“正确”的定义不同，道德判断可能没有唯一答案。
 - Developer A may think it is right that children can freely access horror stories online, relying on parental control, but developer B may think there needs to be inbuilt automatic filters.

强迫 人可能被外部压力迫使去做不道德的事情；

- **Coercion**. People may be deliberately forced to do something unethical, but in modern world it will be more subtle: **peer pressure, or managerial pressure, withholding of benefits, blackmail** and so on. 在现代社会，强迫往往是间接的、心理层面的，例如：
 - 同侪压力 (peer pressure): 同事都这样做，你不做就被孤立；
 - 管理压力 (managerial pressure): 上司要求造假、隐瞒数据；
 - 利益控制 (withholding of benefits): 不给升职、扣奖金；
 - 敲诈勒索 (blackmail): 掌握隐私信息威胁服从。

人性的复杂：人类行为不能简单划分为“好人”与“坏人”

- **People are more complex than simply “good” and “evil”!**
 - Watch The Milgram Experiment, Derren Brown: How “ordinary people” can be compelled to harm others:
<https://www.youtube.com/watch?v=Xxq4QtK3j0Y>

🧠 三、与前页内容的整合分析 (为什么人会不道德)

类型	原因	表现形式	伦理理论对应
无知 (Ignorance)	缺乏专业知识	造成可避免的错误、bug	专业能力义务 (Competence)
疏忽 (Negligence)	不负责任/不测试	数据泄漏、事故	义务论 (应尽职责)
故意 (Intentional Misconduct)	利益或欺骗	撒谎、隐瞒风险	功利主义/义务论
相对主义 (Relativism)	标准差异	冲突决策、文化矛盾	文化相对主义
强迫 (Coercion)	权威压力	服从命令、造假	关怀伦理/契约论
人性复杂性 (Human complexity)	环境影响	盲从权威、群体效应	社会心理伦理学

• Analyze a scenario using the BCS Code Conduct.

编号	原文	P7的知识点总结	中文含义	实践要求
1	Contribute positively to society and human well-being.		积极促进社会与人类福祉。	设计的系统应帮助社会发展、提升幸福感、避免造成伤害。开发者应考虑产品的社会后果，而不仅是功能实现。
2	Be honest, transparent and trustworthy.		诚实、透明、值得信赖。	不隐瞒缺陷、不夸大性能；算法逻辑和数据使用应可解释、可追溯。
3	Maintain professional competence for yourself and other coworkers.		保持自身与同事的专业能力。	持续学习新技术与法规；遵循行业最佳实践；避免因无知或懈怠造成错误。
4	Respect data privacy and intellectual property rights.		尊重数据隐私与知识产权。	按法律与伦理规范处理用户数据；不滥用他人代码或信息。
5	Be fair, do not discriminate, and take action against discrimination.		公平公正，不歧视，并积极反歧视。	设计算法和系统时要防止偏见（如性别、年龄、种族、心理健康），并主动识别潜在的不公。

P7可以补充的伦理原则			
6	Accountability (问责)		开发者与机构必须对技术后果负责（尤其是 AI 与算法系统）。
7	Safety & Reliability (安全与可靠性)		系统应经过严格测试与验证，确保不会造成意外风险。
8	Explainability (可解释性)		算法决策应能被人类理解与追溯。
9	Sustainability (可持续性)		技术应长期有益，不制造环境或社会负担。
10	Non-maleficence (不伤害原则)		来自医学伦理的核心理念——“首先，不造成伤害”。